AIDAL-ARG at Daleel 2026: Hybrid and Multitask Approaches for Arabic Argument Mining

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Abstract

This paper presents our systems for the Daleel 2026 Shared Task. We explore multitask learning, domain-specific ensembles, and a two-stage BIO segmentation and classification pipeline with LLM-based disambiguation, alongside classical machine-learning, Transformer, and LLM baselines. Results show that supervised Transformers outperform standalone LLM prompting, while LLMs are more effective for selectively resolving ambiguous predictions. The multitask model achieves the best test performance on Task 1, with an F1 score of 0.599, while the ensemble multitask model reaches 0.683 F1 on Task 2. Our results show the potential of LLM integration within Transformer systems for Arabic argumentative discourse mining.

1 Introduction

Argument Mining (AM) has advanced considerably with neural and LLM-based methods, but research remains largely centered on high-resource languages such as English, while Arabic suffers from limited annotated resources and additional linguistic challenges, including dialectal variation and code-switching. These limitations motivate the development of dedicated resources and methods for Arabic argumentative discourse mining. The Daleel 2026 Shared Task (Nabhani et al., 2026) addresses this gap through two subtasks across editorials and debates, covering six Argumentative Discourse Unit (ADU) categories: Common Ground (CO), Assumption (AS), Testimony (TE), Statistics (ST), Anecdote (AN), and Other (OT). Task 1 focuses on ADU classification, while Task 2 focuses on identifying the corresponding spans within a paragraph.

This paper presents our submissions to the Closed Track for both subtasks. We compare a pipeline that combines BIO-based segmentation, a transformer classifier, and LLM-based disambiguation of uncertain predictions with multitask architectures that jointly learn segmentation and classification. Our submissions ranked 7th out of 19 teams in Task 1 and 8th out of 14 teams in Task 2. The results show that the pipeline achieves the best performance on the development set, whereas the multitask approach achieves stronger performance on the official test set, particularly for Task 1. Finally, our analysis identifies the confusion between Common Ground and Assumption as one of the main remaining challenges, highlighting the semantic similarity between these categories. The code is publicly available on GitHub.[1]

2 Background

AM is a well-established area in computational linguistics concerned with the automatic identification and classification of argumentative structures (Berghegger et al., 2025; Cabrio and Villata, 2018; Lawrence and Reed, 2019). A fundamental step is the identification of Argumentative Discourse Units (ADUs), the minimal text spans serving distinct argumentative functions. While most AM research has focused on high-resource languages, recent efforts have begun developing resources and methods specifically for Arabic argumentative discourse. Khader et al. (2024) introduced Munazarat 1.0, a corpus of Arabic competitive debates designed to support argumentation mining, followed by work on dedicated annotation schemes for Arabic argumentative debates (Al-Sharafi et al., 2025) and neural classification of argument elements and styles (Al-Zawqari et al., 2025). More recently, Nabhani and Khatib (2025) studied Arabic argument component identification as a BIO sequence-labeling task, exploring cross-lingual transfer, translation-based training, and LLM prompting to address the scarcity of native annotated resources. Beyond ar-

[1] https://github.com/KarlaDSJ/Daleel2026-AIDAL-ARG

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1349–1357
October 28-29, 2026 ©2026 Association for Computational Linguistics

gument mining, BIO-based segmentation has also proved effective for Arabic and multilingual discourse segmentation (Keskes et al., 2013; Pujol et al., 2025).

Recent work has combined conventional models or rules with LLMs used selectively for ambiguous cases. Grecu et al. (2026), for example, use structural heuristics with a zero-shot LLM fallback in the ArgMining 2026 shared task, reducing computational cost while maintaining competitive performance. Other studies have explored LLM-based embeddings for unsupervised segmentation (Krassovitskiy et al., 2025) and generative approaches to AM (Cabessa et al., 2025).

Argument mining commonly involves both identifying argumentative unit boundaries and determining their argumentative functions. These stages have traditionally been modeled separately, with argument-unit segmentation formulated as sequence labeling (Ajjour et al., 2017; Petasis, 2019). More recent work has explored joint and multitask architectures to exploit dependencies between related AM subtasks. For example, Habernal et al. (2024) jointly predict argument type and speaker role in legal texts using shared Transformer representations. Such joint learning allows related tasks to benefit from a common contextual representation rather than being optimized independently. This motivates our comparison between a pipeline that separates ADU segmentation and classification and a multitask architecture that learns both tasks through a shared Arabic Transformer encoder.

3 System Overview

This section describes our three best-performing systems: a multitask architecture for joint ADU segmentation and classification, its ensemble extension, and a pipeline combining separate segmentation and classification with ensemble disambiguation. For all three approaches, we focus on span identification and classification in Task 2. The multi-label paragraph output required for Task 1 is then derived by collecting the unique labels of all predicted spans within each paragraph. We also explored alternative approaches, with the best approaches reported in Section 5 and the remaining experiments summarized in Appendix A.

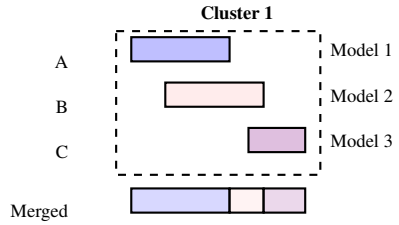


Figure 1: Example of character-level span clustering given 3 sets of predicted spans extracted from the same paragraph. Overlapping spans form a connected cluster and are merged using the minimum start and maximum end offsets.

3.1 Multitask

Our base model jointly learns ADU segmentation and classification using a shared Arabic transformer encoder: *CAMeL-Bert-mix* (Inoue et al., 2021). Two task-specific heads operate on the shared contextualized representations.

The segmentation head predicts BIO tags (B-ADU, I-ADU, O) through a token-level linear classifier. The classification head pools the hidden representations of each span using mean pooling and feeds the resulting vector into a feed-forward network to predict one of seven labels (CO, AS, TE, ST, AN, OT, or NT, the NT label is explained in Section 4).

During training, the classification head receives gold ADU boundaries (teacher forcing), while the segmentation head is optimized independently on the BIO tagging objective. Both tasks are trained jointly using a weighted sum of their focal losses.

3.2 Ensemble multitask

Our second system combines three multitask models trained on different subsets of the training data: one trained on the complete set, one using only debate documents, and one using only editorial documents. Each model is trained independently using the same architecture.

To compare and combine the arguments identified by each model we introduce a span-agreement mechanism. Given the 3 sets of predicted spans extracted from the same paragraph, we merge them based on their character-level overlap. First, all spans from all prediction models are represented as nodes. Two nodes are connected whenever their spans overlap.

We then compute connected components over this overlap relation. In this way, overlapping spans are grouped into clusters, including cases in which

two spans do not directly overlap but are connected through an intermediate overlapping span, for example the case of *A* and *C* in Figure 1. Only clusters containing predictions from all models are retained and merged into a single span whose boundaries correspond to the minimum start offset and maximum end offset among its members.

When all models assign the same label, it is kept as the final prediction. Otherwise, the disagreement is resolved using *CohereLabs/aya-expanse-8b* (Dang et al., 2024), which receives the candidate ADU, its paragraph context, and the set of possible labels, and selects the most appropriate argumentative category.

3.3 BIO segmentation + ensemble

We extend the standard BIO segmentation by assigning label-specific BIO tags (e.g., B-AS, I-AS, B-AN), using the same architecture as the multitask model. The predicted spans are then classified with *Arabertv02-twitter* (Antoun et al., 2020), which provides class probabilities used to identify sensitive cases, i.e., instances where the confidence difference between the top prediction and competing labels is below 0.1, or where the predicted label disagrees with the BIO tag. These sensitive cases are translated into English with *CohereLabs/aya-expanse-8b* and resolved by *Meta-Llama-3.1-8B-Instruct* (Dubey et al., 2024), which selects the final label using the ADU, its paragraph context, and the candidate labels.

4 Experimental Setup

During an initial inspection of the dataset, we analyzed the class distribution across the training, development, and test sets. As shown in Table 1, the distribution is largely consistent across splits, while also revealing a pronounced class imbalance: AS is the dominant category, accounting for approximately 41–42% of instances, whereas ST and CO each represent less than 4%, with OT, AN, and TE occupying intermediate proportions.

Split	AS	AN	ST	TE	CO	OT	Total
Train	42.4	16.7	2.6	14.7	3.3	20.4	1090
Dev	40.8	17.7	2.3	19.7	3.1	16.4	385
Test	41.2	12.0	3.5	18.6	2.7	22.1	376

Table 1: Class distribution (%) across the train, dev, and test sets.

We further observed that some paragraphs contain no gold ADU spans and are represented as *[]* in the original annotations. For training, we introduced an auxiliary *Nothing* (*NT*) label for these instances, allowing the model to learn that a paragraph may contain no ADU span. At prediction time, *NT* instances are converted back to *[]* to preserve the original shared task format.

Approach	Prec	Rec	F1
Dev set			
BIO seg. + ensemble	0.657	0.703	**0.679**
Multitask	0.596	0.570	0.575
Ensemble multitask	0.645	0.482	0.541
Test set			
BIO seg. + ensemble	0.616	0.584	0.579
Multitask	0.633	0.581	**0.599**
Ensemble multitask	0.642	0.493	0.546

Table 2: Performance of the three submitted systems on the dev and the test set. Results are reported in terms of macro values.

Each paragraph was split into its annotated ADUs (*span_text*), and the official training set was divided into training and validation partitions using an 80/20 *GroupShuffleSplit*[2], ensuring that ADUs from the same paragraph remained in the same partition. Importantly, this division is applied only for training. The results reported in Section 5 are evaluated on the official Dev and Test sets provided by the shared task organizers.

The first two approaches use 0.1 dropout, mean pooling for span representations, three training epochs, a learning rate of 2×10^{-5}, and equally weighted BIO and classification losses (1.0 each). For the third approach, both models are trained for five epochs with the same learning rate and a weight decay of 0.01, using batch sizes of 8 for segmentation and 16 for classification.

We use the official evaluation code provided by the organizers. Task 1 is evaluated using micro, macro, and weighted precision, recall, and F1, while Task 2 uses a span-overlap metric that assigns partial precision and recall according to the overlap between predicted and gold spans.

5 Results

This section presents the results and analysis for Tasks 1 and 2 in the Closed Track using the approaches described in Section 3. It is important to note that the Multitask and Ensemble Multitask systems were developed after the development phase.

[2] sklearn.model_selection.GroupShuffleSplit

Gold (ID 282): CO [90–193] , translated text: *"Climate change is something all countries agree on, whether they are first-world, second-world, or even third-world countries."* , and
Multitask: AS [90–193 +2]
Ensemble multitask: AS [90–193 +2]
BIO seg. + ensemble: AS [90–193]

Figure 2: Translated example of the CO–AS confusion. All systems identify the span (BIO seg. + ensemble matches the gold boundaries exactly) but classify it as AS instead of CO, illustrating the main Task 1 error.

5.1 Task 1

Table 2 gives the macro-averaged precision, recall, and F1 of our three submitted approaches on the dev dataset (see Appendix B.1 for information about our experiments with additional approaches).

The BIO seg. + ensemble obtained the best performance on the dev set, outperforming both multitask variants. However, this did not transfer to the test set: its performance dropped noticeably, while the multitask model generalized better, achieving the highest test F1. The ensemble multitask achieved the highest test precision but substantially lower recall, limiting its F1.

The results provide some indication that multitask learning may improve generalization; however, this observation is based on a single comparison, and the differences in Task 2 are relatively small.

Table 3 reports per-class F1 scores for the three submitted systems. AS consistently achieves the highest F1 scores (> 0.85), whereas none of the systems correctly predicts CO. Since ST has fewer dev instances but substantially higher scores, data scarcity alone cannot explain this difficulty, suggesting possible semantic overlap between CO and AS. Figure 2 illustrates such a case.

Approach	AS	AN	ST	TE	CO	OT
Dev set						
Support	157	68	9	76	12	63
BIO seg. + ensemble	0.89	**0.66**	**0.87**	0.82	0.00	**0.67**
Multitask	**0.91**	0.59	0.53	0.78	0.00	0.62
Ensemble multitask	0.90	0.52	0.53	0.68	0.00	0.60
Test set						
Support	155	45	13	70	10	83
BIO seg. + ensemble	**0.94**	**0.50**	**0.76**	0.57	0.00	0.70
Multitask	0.90	0.48	0.72	**0.67**	0.00	**0.79**
Ensemble multitask	0.92	0.41	0.72	0.50	0.00	0.69

Table 3: Per-class F1 scores for the three submitted systems on the dev and test sets.

On the test set, BIO seg. + ensemble shows the sharpest degradation, particularly on TE ($0.82 \rightarrow 0.57$), while Multitask remains more consistent, with OT F1 increasing from 0.62 to 0.79. This reinforces the earlier finding that jointly learning segmentation and classification generalizes better than combining independently trained components.

Table 4 summarizes the most frequent sensitive pairs identified by the ensembles. AS dominates the sensitive cases across both systems, with AS–OT being the most frequent pair on the test set, highlighting the difficulty of distinguishing assumptions from miscellaneous statements. On the dev set, BIO seg. + ensemble also shows frequent NT-related pairs, reflecting instances without an ADU. In contrast, the ensemble multitask exhibits a more stable AS–TE confusion, which remains among its most frequent pairs across both datasets.

System	Sensitive pair	Freq.
Dev set		
BIO seg. + ensemble	NT–OT	34.7%
	AS–NT	32.1%
	AS–OT	29.8%
Ensemble multitask	AS–TE	41.1%
	AN–AS	32.3%
	AS–OT	23.5%
Test dataset		
BIO seg. + ensemble	AS–OT	45.4%
	AN–AS	26.4%
	AS–TE	13.2%
Ensemble multitask	AS–OT	55.5%
	AN–AS	25.5%
	AS–TE	22.0%

Table 4: Most frequent sensitive label pairs for the BIO + ensemble and ensemble multitask systems on the dev and test sets. Frequencies are computed over sensitive cases.

5.2 Task 2

Table 5 reports the performance obtained on Task 2 for our three submissions (see Appendix B.2 for a discussion of the other approaches).

BIO seg. + ensemble achieved the highest performance on the development set. On the test set, however, the three approaches performed comparably, with the multitask and ensemble multitask models obtaining only slightly higher F1 scores than the BIO segmentation pipeline. This provides limited evidence that joint segmentation and classification may improve robustness.

To better understand the differences observed in Table 5, Table 6 analyzes the quality of the predicted span boundaries on the dev set. The BIO segmentation + ensemble obtains the highest recall in Task 2, consistent with its higher proportion of gold spans matched by the predicted spans.

Approach	Prec	Rec	F1
Dev dataset			
BIO seg. + ensemble	0.609	0.761	**0.677**
Multitask	0.610	0.677	0.642
Ensemble multitask	0.627	0.632	0.629
Test set			
BIO seg. + ensemble	0.724	0.635	0.677
Multitask	0.647	0.712	0.679
Ensemble multitask	–	–	**0.683**

Table 5: Performance comparison of the evaluated approaches for Task 2, using the best-performing Task 1 system as a reference and reporting the three submitted systems on the development and test sets.

In contrast, the multitask model predicts fewer spans but locates the beginning of ADUs more precisely, achieving the highest proportion of exact start boundaries and the smallest average start error.

Finally, the ensemble multitask model detects the fewest gold spans, explaining its lower recall. However, its boundary predictions remain relatively accurate, particularly at the end of ADUs, suggesting that the ensemble favors more conservative but consistent span predictions, Figure 2 shows an example.

Metric	BIO+Ens.	MT	Ens. MT
Matched spans	**81.6**	79.9	72.7
Exact boundaries	25.8	**27.7**	26.5
Exact start	38.5	**45.2**	43.9
Starts earlier	**24.6**	30.1	37.5
Starts later	36.9	24.7	**18.6**
Exact end	68.4	63.0	**62.7**
Ends earlier	14.4	16.0	**10.4**
Ends later	17.2	21.0	**26.9**
Avg. start error	14.6	**12.3**	14.0
Avg. end error	**16.5**	17.8	21.7

Table 6: Comparison of span boundary predictions on the development set. Percentages are computed over matched spans, and average boundary errors are reported in characters.

6 Conclusion

This paper presented our submissions to the closed Daleel 2026 Shared Tasks on Arabic Argumentative Discourse Mining, exploring supervised Transformers, LLM prompting, ensemble, and multitask learning for ADU segmentation and classification. Our results show that Transformers outperform standalone LLM prompting, while LLMs are more effective for resolving ambiguous predictions. Although the two-stage ensemble achieved the strongest development performance, the multitask architecture showed more stable performance on the official test set, particularly for Task 1. Overall, our findings suggest that joint segmentation and classification is a promising direction and that LLMs may be better suited as complementary components within hybrid systems than as standalone classifiers for low-resource Arabic AM.

Acknowledgements

This work is funded by the French National Research Agency (grant ANR-22-CPJ1-0061-01).

Experiments presented in this paper were carried out using the OCCIDATA platform that is administered by IRIT and supported by CNRS (https://occidata.irit.fr).

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A Additional Experiments

We describe all the variants, although we only report the best results of each category.

A.1 Segmentation rule-based

The following experiments use our first segmentation approach to split Arabic paragraphs using three types of linguistic cues: (1) question marks, (2) discourse markers, and (3) argumentative verbs.

Discursive markers: However, yet, nevertheless, on the other hand, because, since, whereas, consequently, therefore, for this reason, for example, despite, although, due to, in addition, furthermore, on the other hand, but, likewise.

Argument verbs: He said, they said, we said, you said, we proved, they assumed, you assumed, you explained, you did not explain, they did not answer.

A.2 Classic ML

As an initial baseline, we compared three classical algorithms: Naive Bayes (NB), Logistic Regression (LR), and Support Vector Machines (SVM) using Arabic *span_text* embeddings. Since NB achieved the best validation performance, it was selected for the remaining experiments in this category. We evaluated five embedding models: *Omartificial-Intelligence-Space/Arabert-all-nli-triplet-Matryoshka*, *Omartificial-Intelligence-Space/Arabic-Triplet-Matryoshka-V2* (Nacar and Koubaa, 2025), *sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2* (Reimers and Gurevych, 2019), *Omartificial-Intelligence-Space/GATE-AraBert-v1* (Nacar et al., 2025), and *sentence-transformers/paraphrase-multilingual-mpnet-base-v2* (Reimers and Gurevych, 2019), of which the first consistently performed best (only these results are reported in Table 7). We also tested three document-type settings: combined debate+editorial, debate-only, and editorial-only, and, separately, translated the Arabic ADUs into English with *facebook/nllb-200-distilled-600M* (Team et al., 2022) and generated embeddings with *BAAI/bge-m3* (Chen et al., 2024) to train the same NB classifier on both document types.

A.3 Transformers

All Transformer models were fine-tuned for five epochs with a learning rate of 2×10^{-5}, batch size of 16, and weight decay of 0.01. We evaluated the following Arabic-centric models: *CAMeL-Lab/bert-base-arabic-camelbert-mix* (Inoue et al., 2021), *aubmindlab/bert-base-arabertv02*, *aubmindlab/bert-base-arabertv2_all*, *aubmindlab/bert-large-arabertv02*, and *aubmindlab/bert-base-arabertv02-twitter* (Antoun et al., 2020), with the latter achieving the best performance. We also evaluated two English-centric models, *distilroberta-base* (Sanh et al., 2020) and *answerdotai/ModernBERT-base* (Warner et al., 2025), using segments and paragraphs translated with *CohereLabs/aya-expanse-8b*. Finally, we trained separate models per document type (debate and editorial) and combined their predictions into a single output.

A.4 LLMs

Our initial prompting strategy jointly addressed both subtasks, prompting LLMs to extract ADUs and assign their labels. We first used free-text gen-

eration and then constrained the output to JSON format. We evaluated the Arabic-centric models *FreedomIntelligence/AceGPT-13B-chat* (Huang et al., 2024) and *CohereLabs/aya-expanse-8b*, and the English-centric models *meta-llama/Meta-Llama-3.1-8B-Instruct* and *Qwen/Qwen2.5-7B-Instruct* (Team, 2024). Decoding was performed with a maximum of 500 new tokens, *do_sample=False*, a repetition penalty of 1.15, and *no_repeat_ngram_size=4* to reduce repetition and improve span recovery.

A.5 Ensembles

Transformers + prompt sensitive cases

We first obtained class probabilities from the Transformer models (Section A.3). Predictions with a confidence margin above one percentage point were kept unchanged. For the remaining *sensitive cases*, we collected all candidate labels within one percentage point of the top score and prompted an LLM to select the final label using only the definitions of the candidate labels, thereby limiting the risk of hallucination.

We evaluated this strategy using predictions from *aubmindlab/bert-base-arabertv02-twitter* and *CAMeL-Lab/bert-base-arabic-camelbert-mix*, with *Meta-Llama-3.1-8B-Instruct* and *Qwen2.5-7B-Instruct* as the disambiguation LLMs.

Prompts were written in English using translated ADUs, with an additional Arabic-language experiment using Llama. Decoding followed the same configuration as in Section A.4. All prompts are available at `https://github.com/KarlaDSJ/Daleel2026-AIDAL-ARG/blob/main/experiments/run.ipynb`.

One model per label

We trained one binary classifier per label (target label = 1, all others = 0). At inference, if exactly one classifier predicted as positive a given instance, that prediction was accepted directly. Otherwise, when multiple classifiers predicted as positive the same instance, the sensitive case prompt was used with Llama to select the final label. As a baseline, we replaced the LLM decision with a random choice among the candidate labels.

Argument generation with positive and negative reasoning

Extending the sensitive-case strategy but restricted to predictions from *aubmindlab/bert-base-arabertv02-twitter* (the best-performing model), we used a multi-step prompt: for each candidate label, the LLM first generated an argument for and an argument against that label (given the ADU, its full paragraph as context, the target label, and the competing labels), and a final prompt then selected the most appropriate label based on all generated arguments.

LLMs with different temperatures

The LLM assigned a label to each segment at temperatures of 0.5, 0.7, and 0.9, and the resulting predictions were combined following the same procedure as in *One model per label*.

B Results of other approaches

B.1 Task 1

Table 7 gives the macro-averaged precision, recall, and F1 of the best evaluated approaches on the dev dataset.

Approach	Prec	Rec	F1
Dev set			
Classic ML			
NB all	0.519	0.622	0.552
NB debate	0.419	0.550	0.463
NB editorial	0.480	0.456	0.442
NB all (eng/bge-m3)	0.548	0.594	**0.560**
Transformers			
camelbert-mix	0.627	0.583	0.587
arabertv02 large	0.452	0.474	0.462
ModernBERT base	0.589	0.562	0.565
distilroberta-base	0.566	0.598	0.574
arabertv02 base	0.627	0.568	0.575
arabertv02-twitter	0.592	0.628	0.600
arabertv02-twitter × type	0.607	0.664	**0.603**
LLMs			
aya-expanse JSON	0.440	0.589	0.413
aya-expanse no format	0.380	0.587	0.351
AceGPT JSON	0.323	0.514	0.363
Qwen JSON	0.406	0.595	0.438
Llama JSON	0.612	0.486	**0.444**
Ensembles			
camelbert-mix + Llama (sensitive case)	0.579	0.572	0.564
Llama self-consistency (temp.)	0.456	0.523	0.411
arabertv02-twitter based:			
+ Llama (sensitive case arabic)	0.645	0.653	0.637
+ Qwen disambiguation	0.630	0.652	0.621
× doc-type + Llama disamb.	0.561	0.647	0.589
one-vs-rest + Llama tie-break	0.659	0.582	0.563
one-vs-rest + random tie-break	0.666	0.623	0.561
+ Llama (args for/against)	0.568	0.645	0.603
+ Llama (sensitive case)	0.686	0.657	**0.641**

Table 7: Performance of other approaches on the official dev set. Results are reported in terms of macro precision, recall, and F1.

Transformer models consistently outperformed Classic ML and prompting approaches, with *aubmindlab/bert-base-arabertv02-twitter* as the strongest individual classifier (0.600), only

marginally improved by training separate debate/editorial variants (0.603). Other pretrained models, particularly *arabertv02 large* (0.462), performed considerably worse. For Classic ML, translating ADUs into English before embedding (*NB all (eng/bge-m3)*) outperformed all Arabic-embedding variants, while splitting NB by document type hurt performance (debate: 0.463, editorial: 0.442) relative to the combined model (0.552). Prompting-based LLMs remained far less competitive than supervised approaches, with *Llama JSON* as the best of this family (0.444).

Among ensemble strategies, combining *arabertv02-twitter* with Llama as a sensitive-case disambiguation tie-breaker achieved the best result (0.641). Prompting the disambiguation step in Arabic yielded a nearly identical score (0.637), indicating the prompt language had little impact, while using Qwen instead of Llama for disambiguation was slightly less effective (0.621). Other ensemble variants, one-vs-rest with Llama or random tie-breaking, doc-type-split ensembling, and the for/against argument-generation strategy, all underperformed relative to the simple sensitive-case approach, and self-consistency across decoding temperatures was the weakest ensemble overall (0.411), suggesting that disagreement among independent classifiers is a more useful uncertainty signal than sampling variance from a single model.

B.2 Task 2

Table 8 reports Task 2 performance for the best model of each approach evaluated in Task 1. Results are consistent with Task 1: the Transformer classifier (*arabertv02-twitter* × *type*, F1=0.598) outperformed both Classic ML (0.494) and the standalone LLM (0.577), and the proposed ensemble further improved performance, raising the F1 score to 0.608, driven primarily by a gain in recall (0.647 vs. 0.563 for the Transformer alone).

Approach	Prec	Rec	F1
Dev dataset			
NB all (eng)	0.460	0.532	0.494
arabertv02-twitter × type	0.563	0.563	0.598
Llama JSON	0.523	0.478	0.577
arabertv02-twitter + Llama	0.573	0.647	**0.608**

Table 8: Performance comparison of the evaluated approaches for Task 2, using the best-performing Task 1 system as a reference.